Agentic Engineering

Realita, Strategi, dan Praktik
untuk Tim Produk & Bisnis Digital

👋 Perkenalan

🔍 **Google aja nama saya** — Zain Fathoni

- 📍 Jember ➡️ Bandung ➡️ 🇸🇬 SG ➡️ Jogja
- 🛠️ Backend ➡️ Manager ➡️ Frontend ➡️ Fullstack
- 🤖 Agentic Engineering Practitioner
- 🤝 Talk ini untuk semua: dev, founder, marketer, dan produk

Agenda

1. **The Reality of AI-Assisted Development**
2. **Coverage & Control**
3. **Prompting & Interaction Strategy**
4. **Structured AI Coding**
5. **Maintaining Craftsmanship**
6. **Live Demo Results** 👀

🎬 Live Demo Setup

Jalankan Claude Code di terminal, lalu lanjut presentasi

```
claude "Add filter by category (Conferences /
 Podcasts / Meetups) to the /talks page.
 Reference: filter pattern in /blog. Create a PR."
```

⏳ Claude Code bekerja di background...

1) The Reality of AI-Assisted Development

🤖 Apa itu Agentic Engineering?

> "Agentic engineering: developing software with the assistance of coding agents that can write **and execute** code."
> — Simon Willison

- 👓 Agent bekerja dalam loop: tulis kode → jalankan → iterasi
- 🧠 Tugas kita: **tentukan** masalah, **verifikasi** hasil, **iterasi**
- ❌ Bukan "vibe coding" — bukan asal generate dan lupakan
- ✅ Ini yang kita lakukan hari ini dengan Claude Code

🤖 Peran AI di Setiap Fase Kerja Digital

Fase	Tools AI	Peran
🎨 **Design**	v0, Figma AI	Prototyping & landing page
📝 **Content**	ChatGPT, Claude	Naskah, kerangka, dokumentasi
💻 **Coding**	Claude Code, Copilot	Implementasi
🧪 **Testing**	Playwright MCP	Pengujian otomatis
🤖 **Agents**	OpenClaw, Codex CLI	Task end-to-end via chat

✨ Hype vs Reality

> "AI will replace developers"

- ✅ AI **mempercepat** pekerjaan repetitif & eksplorasi solusi
- ❌ AI **belum bisa** menggantikan pemahaman arsitektur
- ❌ AI **sering salah** di edge case dan konteks bisnis
- ❌ AI **butuh arahan** jelas dari manusia yang paham konteks

⚠️ Jebakan Umum

- 🙈 **Copy-paste tanpa review** → bug tersembunyi
- 🍝 **Tanpa arsitektur** → sulit di-maintain
- 🔓 **Titik buta keamanan** → vulnerability lolos
- 💣 **Over-confidence** → "AI yang bikin, pasti benar"

👥 Siapa yang Ada di Ruangan Ini?

Snapshot Peserta EDC Jogja

- 🏢 **Mayoritas sektor TIK**, mayoritas dari DIY
- 👨‍💻 Terbesar: **Software & Web Developer**
- 💼 Juga ada **founder, business owner, marketer, mahasiswa**
- 🌱 Usia dominan: **22-35 tahun**

👉 **Fokus: AI untuk kolaborasi lintas fungsi, bukan cuma ngoding cepat**

🗣️ Cerita Nyata: Yang Berhasil

- Migrasi Remix → React Router v7 — AI tangani 80% boilerplate
- Refactor styling ke Tailwind — konsisten di 30+ komponen
- Personal AI army (Project Transformers) — 15 agen khusus, masing-masing punya domain sendiri, context tidak bocor antar agen
- Slide ini: draft pertama via Claude Web + Wheeljack (OpenClaw)

Chat Cowork Code

Draft talk slides with live demo

Share

Updated todo list, loaded tools, created a file

Now let me commit and push to the feature branch.

Ran 3 commands, updated todo list

Done! Here's what I created:

`public/ai-assisted-development.md` — 27 slides in your Marp format covering all 5 sections from the ToR:

1. **The Reality of AI-Assisted Development** — AI's role in each dev phase, hype vs reality, common pitfalls
2. **Coverage & Control** — the spectrum from assistive to autonomous, framework for when to use what level
3. **Prompting & Interaction Strategy** — context engineering, role-based interaction (architect/implementer/reviewer), iterative prompting, CLAUDE.md
4. **Structured AI Coding** — requirement → flow → architecture → implementation, Kiro specs
5. **Maintaining Craftsmanship** — code review, security awareness, system thinking, practical do's & don'ts

Live demo framing: The slides include a setup at the beginning where you prompt AI to do a task on the repo, and a "Demo Results" section at the end to check what happened — perfectly demonstrating AI-assisted development in action during the talk itself.

The slides reuse your existing assets (`zain-1280.jpeg`, `prompt-vs-context-engineering.webp`, `ai-agency-human-agency.webp`, `iron-man.png`) and follow the same Marp conventions as your other decks (gaia theme, emoji markers, speaker notes in HTML comments, bilingual Indonesian/English).

1 step

main ← claude/draft-talk-slides-4NE5J +1384 -0 View PR

Reply...

Opus 4.6 1M

Web

ZF

Telegram @ Zain

All Chats · Autobots · Personal · Communities · Alumni · Gaming · BTC · Unread · Other Comms · News · COVID-19

Chats

Search (⌘K)

The Ark — Tue — Jazz: ⚠️ Something went wrong while processi...

Optimus Prime — 08:02 — Autobots Board — Thu Apr 9, 8:01 AM Cards by Column (15 total) 🟡 In Progre...

Red Alert — 08:01 — This request was blocked by the application-layer security filter because the same cron pr...

Wheeljack — Wed — Already done — the HTML and PDF were regenerated and included in that same comm...

Prowl — Wed — Model reset to default (opus (anthropic/claude-opus-4-6)).

Hound — Wed — OpenClaw 2026.4.5 (3e72c03) 🧠 Model: anthropic/claude-sonnet-4-6 · 🔑 token (ant...

Bumblebee — Tue — ⚠️ WhatsApp 499 loop — again (Apr 7) This time it cycled every ~30 min all night (01:14–...

Rewind — Sun — Recommendation: The draft needs honest revision. The infrastructure stories (dispatche...

Perceptor — Sat — This is a textbook example of what Dalio describes as the late-stage dynamics of a de...

Jazz — Fri — Approval required. Run: /approve ff9ee8fb-ee63-4a65-8a31-f08917940c...

Arcee — Fri — Approval required. Run: /approve ff9ee8fb-ee63-4a65-8a31-f08917940c...

Alpha Trion — Fri — Approval required. Run: /approve ff9ee8fb-ee63-4a65-8a31-f08917940c...

Bluestreak — Fri — Approval required. Run: /approve ff9ee8fb-ee63-4a65-8a31-f08917940c...

Zavi 🐝 Bumblebee — 31/03/26 — Bumblebee: Usage: /approve <id> allow-once|allow-alway...

Webpage Bot — 30/03/26 — https://ark.zainf.dev/viz/

BotFather — 13/05/17 — Done! Congratulations on your new bot. You will find it at t.me/BukaTila... OPEN

Wheeljack
bot

1. :robot: AI-Assisted Development Practitioner → :robot: Agentic Engineering Practitioner + weave Simon Willison's definition into relevant slides
2. Update the live demo slide to describe the talks filter task as the demo goal (instead of the vague "buatkan slide" prompt)
3. Fix overflowing slides — split/trim the flagged ones (Peran AI table, Cerita Nyata, Coverage/Control)
4. Remove or replace the invisible `ai-agency-human-agency.webp` bg image
5. Add OpenClaw + your recent blog context (arms race, personal AI army) into the content

You will do live on stage:

- Run the actual OpenClaw/Claude Code session to implement the talks filter on `zainf.dev`

Shall I proceed with just the slides changes?

Zain Fathoni 16.13
Yes

Wheeljack 16.13
On it.

Zain Fathoni 16.13
And don't forget to regenerate the HTML & PDF files

Wheeljack 16.13
Good, full picture. Now applying all 5 changes in one pass:

Good — "talks filter" is there, just the check was case-sensitive. Now regenerate HTML/PDF and commit: 16.15

Done. Commit `8e01413` pushed to the same PR branch. 16.15

Here's what got addressed:

copy

```
| Comment                    | What I did
|
| -------------------------- |
------------------------------------------------------
-------------------------- |
| "Use Agentic Engineering"  | Renamed title, bio
bullet, and added a definition slide quoting Simon
Willison   |
| "Live demo has no value"   | Replaced with the
real talks filter task — prompt goes to Wheeljack
via Telegram |
| "Outdated, no OpenClaw"    | Added OpenClaw to the
tools table, demo flow, and demo results slide
|
| "Cerita Nyata overflows"   | Split into two
slides: Yang Berhasil / Yang Gagal
|
| "Coverage image invisible" | Replaced with a 2×2
Coverage/Control matrix table
|
| "Outdated content"         | Referenced "Project
Transformers" and "Arms Race di Kantor" blog posts
|
| "Simon Willison article"   | Cited in the
definition slide + references section
|
```

The only thing **not** implemented is the actual talks filter feature — that's the live demo you'll run on stage. 🎤

Already done — the HTML and PDF were regenerated and included in that same commit `8e01413`. Both files are already in the PR. ✅ 16.16

Write a message...

🗣️ Cerita Nyata: Yang Gagal

- AI generate test yang lolos tapi **tidak test apa-apa** (assertion kosong)
- AI "fix" bug di satu tempat, tapi **break 3 tempat lain**
- AI bikin copy meyakinkan, tapi **nada dan audiens-nya meleset**
- Claire Vo (LaunchDarkly): OpenClaw **hapus family calendar** di percobaan pertama
- PR storm: arms race di kantor, **kualitas review turun** jika tidak disiplin

💡 Jadi Gimana?

- 🛡️ **Trust, but Verify** — selalu review
- 🧪 **Automate checks** — biarkan mesin bantu verifikasi
- 📋 **Mulai dari spec** — tujuan dulu, baru eksekusi
- 🧠 **Pahami output-nya** — jangan jadi penumpang

2) Coverage & Control

📊 Coverage vs Control

	Coverage Rendah	Coverage Tinggi
Control Tinggi	🐢 Lambat tapi aman	🚀 Ideal
Control Rendah	💤 Tidak efisien	🔥 Chaos

👉 **Tujuan: Coverage Tinggi + Control Tinggi**

🚦 Level Kontrol Penggunaan AI

🟢 **Assistive** — AI menyarankan, manusia memutuskan
Autocomplete, grammar check, inline suggestions

🟡 **Semi-Autonomous** — AI kerjakan, manusia review
Draf naskah, generate function, buat proposal

🔴 **Autonomous** — AI end-to-end, manusia verifikasi akhir
Prototype halaman, generate report

🧭 Framework: Kapan & Seberapa Besar

Situasi	Coverage	Control	Contoh
🎨 Prototyping	⬆️ Tinggi	⬇️ Rendah	Bolt.new, v0
📢 Marketing	⬆️ Tinggi	↔️ Sedang	Draf naskah, rencana konten
💼 Business	↔️ Sedang	⬆️ Tinggi	Riset, ringkasan, proposal
🏗️ Production	↔️ Sedang	⬆️ Tinggi	Claude Code + Review
🔒 Security-critical	⬇️ Rendah	⬆️ Sangat Tinggi	Manual + AI assist

🔑 Anda Tetap Pilot-nya

- 👨‍✈️ AI adalah **co-pilot**, bukan kapten
- 🧠 Pahami keputusan pentingnya
- ✅ Yang bertanggung jawab tetap **Anda**
- ⚠️ Mid-career engineers paling terdampak — bukan junior (Simon Willison, Lenny's Podcast)

3) Prompting & Interaction Strategy

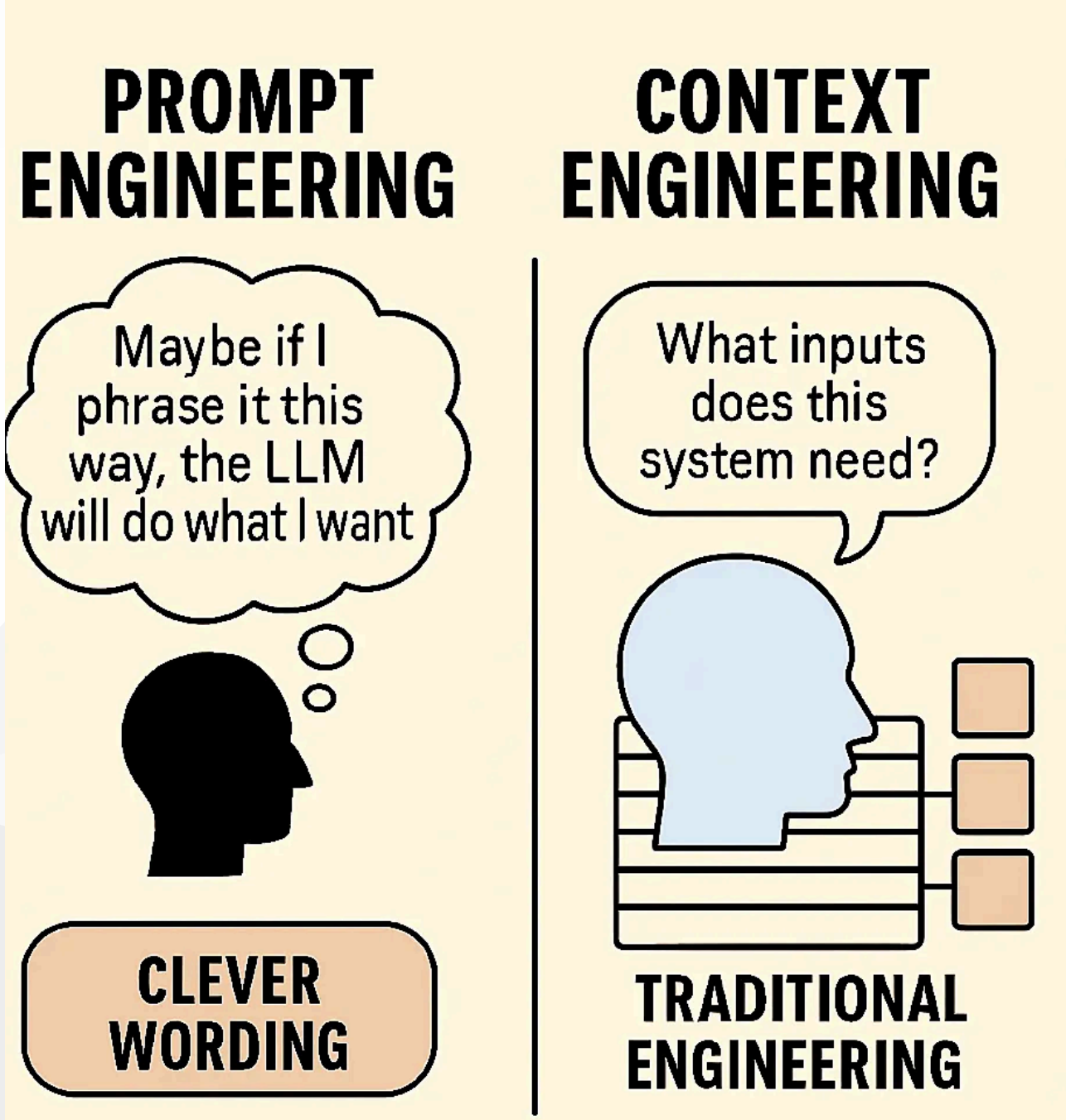
⌨️ Prompting 101

Prompt yang Baik = Hasil yang Baik

- 🎯 **Spesifik** — detail, bukan "buatkan sesuatu"
- 📋 **Kasih konteks** — file, constraint, target audiens
- 🖼️ **Kasih contoh** — tunjukkan format yang dimau
- 🔄 **Iterasi** — perbaiki berdasarkan hasil

🧠 Context Engineering > Prompt Engineering

Bukan hanya **cara bertanya**, tapi **informasi apa yang tersedia**



🧠 5 Jenis Context

- ⌨️ **User Input** — prompt Anda
- 📚 **Knowledge** — docs, riset, transcript
- 💬 **Conversation** — konteks sebelumnya
- 🛠️ **Tool Outputs** — hasil dari tools
- 📂 **Project Memory** — keputusan & artefak yang terus hidup

👉 **Semakin lengkap context, semakin baik output AI**

👥 Role-Based Interaction

Gunakan AI Sesuai Perannya

🏗️ **Architect** — "Desain arsitektur fitur X dengan constraint Y"

💻 **Implementer** — "Implementasikan function X sesuai spec ini"

🔍 **Reviewer** — "Review kode ini untuk security dan performance"

👥 Role-Based Interaction (lanjutan)

Bukan Hanya untuk Developer

📝 **Writing Partner** — "Draft landing page untuk UMKM teknologi di Jogja"

💼 **Business Analyst** — "Ringkas interview customer jadi 5 insight dan 3 aksi"

👉 **Sesuaikan peran AI dengan kebutuhan fungsi Anda**

📂 AI sebagai Second Brain

Insight dari Lenny's Newsletter

- 🧠 **Context is the new interface**
- 📚 Bukan cuma prompt — docs, riset, feedback, keputusan
- 🔍 Context tersimpan rapi → bukan cuma ingat lebih banyak, tapi **berpikir lebih jernih**
- 👥 Berlaku untuk semua fungsi, bukan cuma developer

🔄 Iterasi, Bukan Sekali Jadi

1. 1️⃣ **Prompt awal** → draft pertama
2. 2️⃣ **Review** → identifikasi kekurangan
3. 3️⃣ **Perbaiki** → tambah constraint/context
4. 4️⃣ **Ulangi** sampai kualitas sesuai

💡 **Tips:** Gunakan `plan mode` untuk hemat token

⚡ Prototype dalam Menit, Bukan Minggu

Insight dari Lenny's Newsletter

- 🚀 Prototype interaktif bisa jadi dalam **hitungan menit**
- 💬 Uji ide **sebelum commit build penuh**
- 🎨 Untuk **semua fungsi** — samakan persepsi sebelum eksekusi
- ⚠️ Mulai kecil, iterasi bertahap

👉 **Prototype = alat belajar tercepat**

📜 CLAUDE.md = Autopilot Rules

Konsistensi Output Otomatis

```
# CLAUDE.md
## Git Conventions
- Do not include co-authorship in commit messages

## Architecture
- React Router v7 with file-based routing
- Tailwind CSS for styling
- TypeScript for type safety
```

👉 AI membaca ini dan mengikuti konvensi project

4) Structured AI Coding

🪜 Jangan Langsung Coding!

1. 📋 **Audiens & tujuan** — untuk siapa, dan apa hasil yang dibutuhkan?
2. 🔄 **Alur** — bagaimana user atau tim akan memakainya?
3. 🏗️ **Pendekatan** — bagaimana cara terbaik mengerjakannya?
4. 💻 **Implementasi** — baru mulai eksekusi

🔄 Tanpa vs Dengan Struktur

❌ Tanpa Struktur

```
"Buatkan fitur authentication untuk app saya"
→ Random auth flow, missing edge cases
```

✅ Dengan Struktur

```
Requirement: Login via email/password + OAuth
Flow: Login → Validate → Session → Redirect
Constraints: Rate limiting, CSRF protection
→ AI delivers spec-compliant implementation
```

🧠 Prinsip Dasar

- ⚡ **Cepat tanpa struktur** = utang teknis
- 🛡️ **Output tanpa tinjauan** = risiko
- 📈 **Struktur + AI** = kecepatan berkelanjutan

Mantra:

🎯 Tentukan → 🧪 Uji → 👀 Tinjau

📝 Brief yang Terstruktur

Audiens → Tujuan → Alur → Pendekatan → Eksekusi

- 📝 **kebutuhan** — yang bisa dicek
- 🎨 **desain** — pendekatan dan alur kerja
- 📋 **tugas** — rencana yang bisa dilacak

👉 **AI bekerja lebih baik kalau brief-nya jelas**

5) Maintaining Craftsmanship

💎 Kualitas Tetap Nomor Satu

- 🔍 **Tinjau setiap output** — jangan terima otomatis
- 🧪 **Otomasi** — unit test, integration, E2E
- 📋 **Standar jelas** — checklist, style guide, acceptance criteria
- 👥 **Tinjauan sejawat** — mata manusia tetap penting

🔧 3 Pola Agentic Engineering

dari Simon Willison (Lenny's Podcast)

Pola	Cara Kerja	Contoh Nyata
🔴🟢 **Red/Green TDD**	Tulis test dulu, AI implementasi	Manusia malas TDD — AI tidak
📂 **Templates**	Pola di repo = panduan AI	Pola yang ada, CLAUDE.md
📦 **Hoarding**	Kumpulkan contoh → AI kombinasi untuk kasus baru	Makin banyak pola, makin kreatif AI

🔍 AI untuk Review

- 👃 **Code smell** — duplicated logic, god functions
- ⚠️ **Anti-pattern** — circular deps, tight coupling
- 🐌 **Inefficiency** — N+1 queries, unnecessary re-renders
- 🔒 **Security** — injection, XSS, exposed secrets

🔄 Refactoring dengan AI

- 🎯 **Tentukan target** — pattern baru, bukan "perbaiki semua"
- 🧩 **Batch by scope** — satu modul / satu pattern per iterasi
- 🧪 **Test dulu, refactor kemudian** — pastikan ada jaring pengaman
- ✅ **Contoh nyata:** Refactor styling ke Tailwind — 30+ komponen, konsisten

🔒 Security: Manusia = Final Gatekeeper

- 🛡️ Validasi **vulnerability** — injection, insecure API, supply chain
- ⛔ Jangan percaya begitu saja terhadap AI untuk security
- ⚠️ **Lethal trifecta** — private data + untrusted content + external comms
- 📦 **Supply chain attack** — axios npm dikompromikan Korea Utara
- 🤖 AI boleh **scan**, tapi manusia yang **approve**
- 🔮 Glasswing — Mythos bisa dipakai penyerang & pembela, bersiaplah

⚙️ System Thinking

- 🧩 **Konsistensi** — satu pattern, seluruh produk
- ⛔ Hindari **fragmentasi berlebihan**
- 🔗 **Pahami dampak** setiap perubahan
- 📚 **Docs** — AI bantu generate, manusia review

✅ Do's

- 🔍 **Review** setiap output
- 📋 **Konteks lengkap** — CLAUDE.md, spec, constraint
- 🧪 **Test + evals** otomatis
- 🔄 **Iterasi** bertahap
- 🧠 **Pahami** apa yang di-generate
- 📜 **Rules file** untuk konsistensi

❌ Don'ts

- 🙈 **Copy-paste tanpa baca**
- ❓ **Prompt ambiguous** — garbage in, garbage out
- 👀 **Vibe check doang** — bukan evaluasi
- 💻 **"Works on my machine"** — test di CI
- 🔮 **Berharap sempurna sekali jadi**
- 😴 **Jadi penumpang** di output sendiri
- 🚀 **Mulai tanpa context**

👀 Demo Results

🏆 Waktunya Cek Hasil!

Yang kita minta tadi:

> Filter by category di halaman `/talks` — Conferences, Podcasts, Meetups

👉 Buka terminal → cek status Claude Code
👉 Review PR pakai **GPT-5.4 via Amp Code**
👉 Merge PR → baru zainf.dev/talks terupdate

💡 Key Takeaways

1. 🤖 **Agentic Engineering** — bukan vibe coding, tapi specify → verify → iterate
2. 📊 **Coverage & control** harus seimbang
3. 🧠 **Context > prompt** — termasuk audiens dan tujuan
4. 📋 **Struktur dulu** — audience → goal → flow → approach
5. 💎 **Kualitas tetap nomor satu** — review, test, verify
6. ⚙️ **System thinking** — konsistensi di seluruh produk

🙏 Terima Kasih!

🖥️ https://zainf.dev/agentic-engineering

🔗 https://github.com/zainfathoni/zainf

Ada pertanyaan? ✋

Referensi

- Context Engineering — Addy Osmani
- How to Make Your Tech Stack AI-Friendly — Refactoring · Kiro Specs — AWS
- Lenny's Podcast — How OpenClaw Changed My Life (Claire Vo) · An AI State of the Union (Simon Willison)
- Lenny's Newsletter — *PM second brain with ChatGPT · AI prototyping for PMs*
- What is Agentic Engineering? — Simon Willison
- Arms Race di Kantor · Building My Personal AI Army — Zain Fathoni
- OpenClaw — Agentic engineering platform
- Axios NPM Supply Chain Attack — Google Threat Intelligence · Project Glasswing — Anthropic